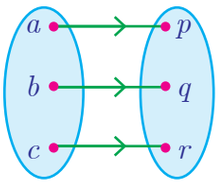
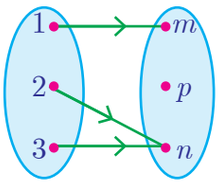
Illustration 9 - Testing for functions

Representation by Arrow diagram



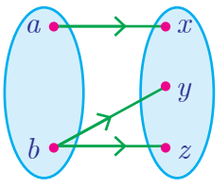
This represents a function. Each input corresponds to a single output.

Fig. 1.13(a)



This represents a function. Each input corresponds to a single output.

Fig. 1.13(b)



This is not a function. One of the input b is associated with two outputs.

Fig. 1.13(c)

Functions play very important role in the understanding of higher ideas in mathematics. They are basic tools to convert from one form to another form. In this sense, functions are widely applied in Engineering Sciences.

Note

The range of a function is a subset of its co-domain.

Example 1.6 Let $X = \{1,2,3,4\}$ and $Y = \{2,4,6,8,10\}$ and R $= \{(1,2),(2,4),(3,6),(4,8)\}$. Show that R is a function and find its domain, co-domain and range?

Solution Pictorial representation of R is given in Fig.1.14. From the diagram, we see that for each $x \in X$, there exists only one $y \in Y$. Thus all elements in X have only one image in Y. Therefore R is a function.

Domain $X = \{1,2,3,4\}$; Co-domain $Y = \{2,4,6,8,10\}$; Range of $f = \{2,4,6,8\}$.

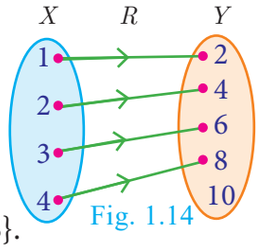


Fig. 1.14

Example 1.7 A relation $f: X \to Y$ is defined by $f(x) = x^2 - 2$ where, $X = \{-2,-1,0,3\}$ and $Y =$ R.

(i) List the elements of f (ii) Is f a function?

Solution $f(x) = x^2 - 2$ where $X = \{-2,-1,0,3\}$

(i) $f(-2) = (-2)^2 - 2 = 2$; $f(-1) = (-1)^2 - 2 = -1$

$f(0) = (0)^2 - 2 = -2$; $f(3) = (3)^2 - 2 = 7$

$\therefore f = \{(-2,2),(-1,-1),(0,-2),(3,7)\}$

(ii) We note that each element in the domain of f has a unique image. Therefore, f is a function.

Thinking Corner

Is the relation representing the association between planets and their respective moons a function?

Example 1.8 If $X = \{-5,1,3,4\}$ and Y $= \{a,b,c\}$, then which of the following relations are functions from X to Y?

1 Relation and Functions.indd 12 15-12-2020 02:50:12 PM